Experiment No. 4

Advanced Data Visualization using Matplotlib and Seaborn

Aim

To create and analyze various data visualizations using Matplotlib and Seaborn in order to identify trends, patterns, correlations and insights from a real-world dataset.

Dataset Used

Superstore Sales Dataset (columns used: Order Date, Region, Category, Sales, Quantity, Discount, Profit). **Note:** the charts and numbers in this report were produced from a synthetic sample with the same column structure as Superstore, so the figures will differ when you run the code on the real file. The code is written for the real dataset; only the file path needs to match yours.

Procedure and Code

Step 1-2: Import libraries and load the dataset

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid", palette="deep")
df = pd.read_csv("Superstore.csv", encoding="latin-1")   # path to your cleaned file
df["Order Date"] = pd.to_datetime(df["Order Date"])
df.info(); df.describe()
```

Step 3: Bar chart - sales by region

```
d = df.groupby("Region")["Sales"].sum().sort_values(ascending=False)
plt.figure(figsize=(6, 4))
ax = sns.barplot(x=d.index, y=d.values, hue=d.index, legend=False)
for p in ax.patches:                        # annotate each bar
    ax.annotate(f"{p.get_height()/1000:.0f}K",
                (p.get_x() + p.get_width()/2, p.get_height()),
                ha="center", va="bottom")
plt.title("Total Sales by Region"); plt.xlabel("Region"); plt.ylabel("Sales ($)")
plt.tight_layout(); plt.show()
```

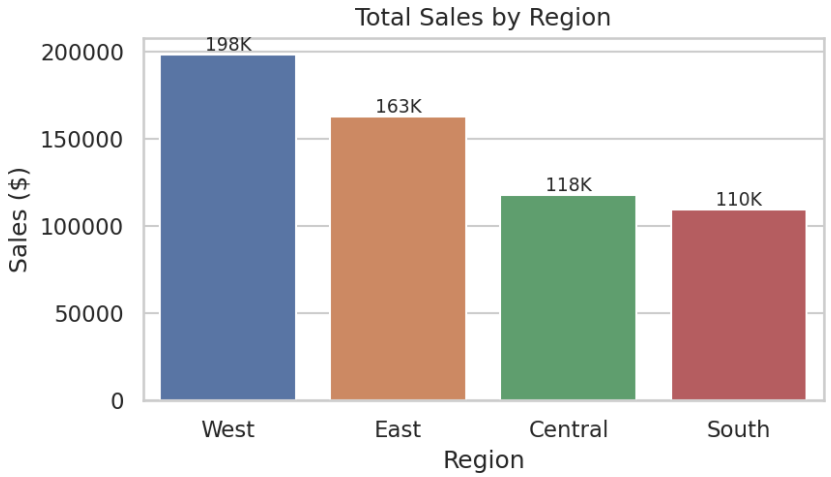


Figure 1: Total sales by region

Step 4: Line chart - monthly sales trend

```
m = df.set_index("Order Date")["Sales"].resample("ME").sum()
plt.figure(figsize=(7, 3.5))
plt.plot(m.index, m.values, marker="o", markersize=3)
plt.annotate("Peak", (m.idxmax(), m.max()), xytext=(0, 12),
             textcoords="offset points", ha="center",
             arrowprops=dict(arrowstyle="->"))
plt.title("Monthly Sales Trend"); plt.xlabel("Month"); plt.ylabel("Sales ($)")
plt.tight_layout(); plt.show()
```

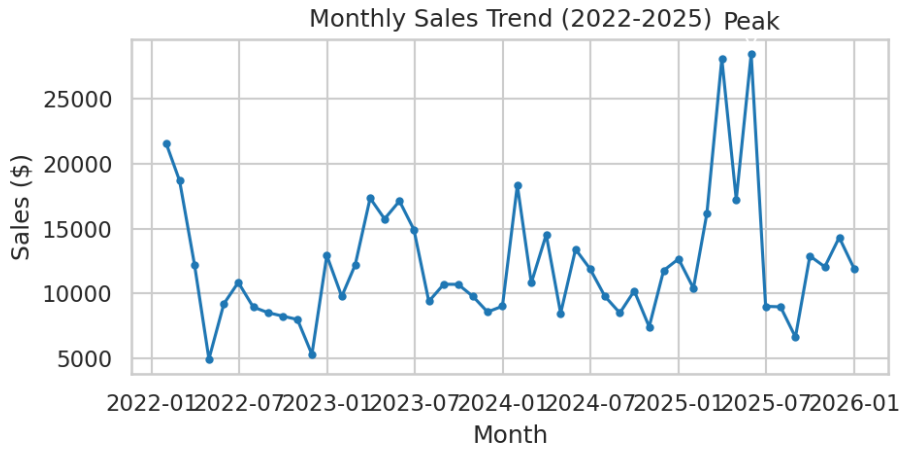


Figure 2: Monthly sales trend

Step 5: Histogram - distribution of sales

```
plt.figure(figsize=(6, 3.5))
sns.histplot(df["Sales"], bins=40, kde=True)
plt.axvline(df["Sales"].mean(), color="red", ls="--", label="Mean")
plt.axvline(df["Sales"].median(), color="green", ls="--", label="Median")
plt.legend(); plt.title("Distribution of Sales")
plt.xlabel("Sales ($)"); plt.ylabel("Frequency"); plt.show()
```

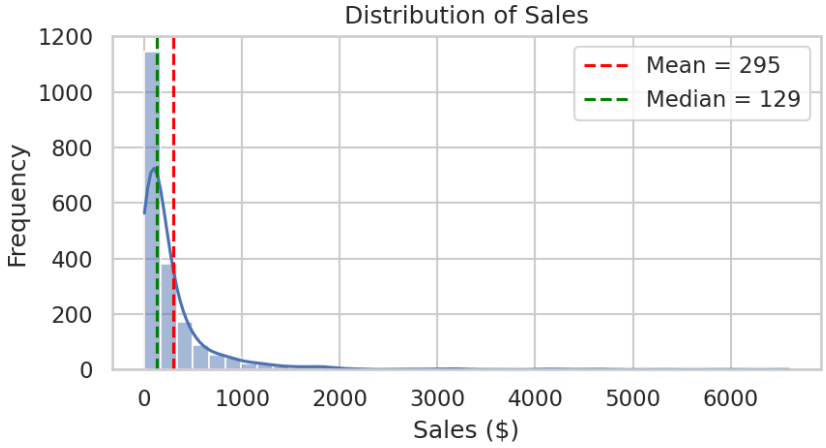


Figure 3: Histogram of sales (right-skewed)

Step 6: Box plot - outliers across categories

```
plt.figure(figsize=(6, 3.8))
sns.boxplot(data=df, x="Category", y="Sales", hue="Category", legend=False)
plt.yscale("log")        # log scale because sales are highly skewed
plt.title("Sales Distribution by Category"); plt.show()
```

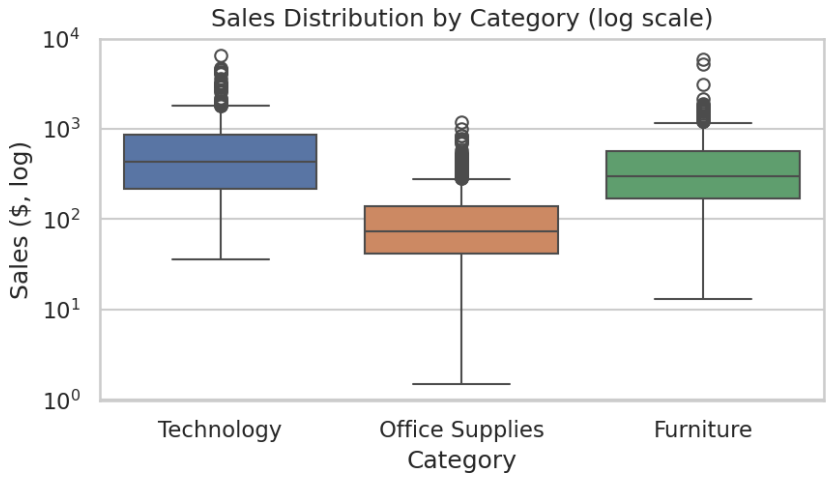


Figure 4: Box plot of sales by category

Step 7: Scatter plot - discount vs profit

```
plt.figure(figsize=(6, 4))
sns.scatterplot(data=df, x="Discount", y="Profit", hue="Category", alpha=0.5)
plt.axhline(0, color="black", lw=0.8)
plt.title("Discount vs Profit"); plt.show()
```

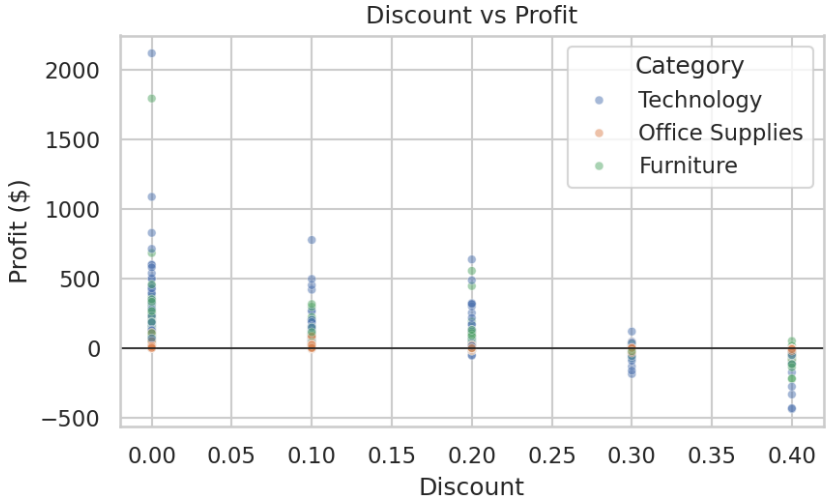


Figure 5: Scatter plot of discount vs profit

Step 8: Correlation heatmap

```
corr = df[["Sales", "Quantity", "Discount", "Profit"]].corr()
plt.figure(figsize=(5.5, 4.2))
sns.heatmap(corr, annot=True, fmt=".2f", cmap="coolwarm", vmin=-1, vmax=1, square=True)
plt.title("Correlation Heatmap"); plt.show()
```

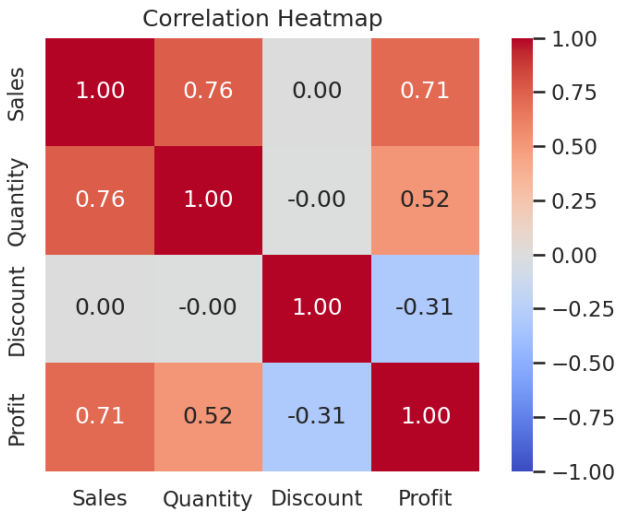


Figure 6: Correlation heatmap

Step 9: Customization

Every chart above uses a descriptive title, labelled axes, a legend where more than one series exists, annotations (bar values, peak marker, mean/median lines), a consistent Seaborn theme and a diverging colour map (coolwarm) for the correlation heatmap.

Step 10: Interpretation and Key Business Insights

- **Bar chart:** West contributes the highest sales and South the lowest. Marketing and stock allocation can be shifted accordingly, and South needs a growth plan.
- **Line chart:** Sales show a gradual upward trend with seasonal peaks. Inventory and staffing should be planned ahead of peak months.
- **Histogram:** The distribution is right-skewed (mean is well above median): most orders are small, a few are very large and drive much of the revenue.
- **Box plot:** Technology and Furniture have higher medians and many high outliers; Office Supplies has low-value orders. Outliers should be checked as genuine bulk orders or data errors.
- **Scatter plot:** Profit falls as discount rises; discounts of 30% or more mostly produce losses. Discount policy should be capped.
- **Heatmap:** Sales and Profit are strongly positively correlated, Discount is negatively correlated with Profit, and Discount has almost no correlation with Sales, suggesting heavy discounting is not buying extra revenue.

Conclusion

Using Matplotlib and Seaborn, bar, line, histogram, box, scatter and heatmap visualizations were created and interpreted. They revealed regional performance, seasonality, skewed distributions, outliers, and the negative effect of high discounts on profit, supporting data-driven business decisions.

Answers to Questions

Q1. Why is data visualization considered an essential component of data analytics?

Raw tables of thousands of rows are hard to interpret, while the human brain processes visuals far faster than numbers. Visualization helps to (a) spot trends, patterns and seasonality, (b) detect outliers and data-quality problems, (c) understand distributions and relationships between variables, (d) communicate findings quickly to non-technical stakeholders, and (e) support faster, evidence-based decisions. It is used both in exploration (EDA) and in final presentation of results.

Q2. Differentiate between Matplotlib and Seaborn. Which library is more suitable for statistical visualizations and why?

Aspect	Matplotlib	Seaborn
Nature	Low-level, base plotting library	High-level library built on top of Matplotlib
Code length	More code for complex plots	Few lines for attractive plots
Statistical features	Manual (compute then plot)	Built-in: KDE, regression lines, confidence intervals, box/violin plots
Default style	Plain	Attractive themes and colour palettes
Pandas integration	Works, but column handling manual	Works directly with DataFrames (data=, x=, y=, hue=)
Customization	Full control over every element	Less fine control, but can be combined with Matplotlib

Conclusion: Seaborn is more suitable for statistical visualizations because it has built-in support for distributions, categorical comparisons, regression and correlation plots with automatic aggregation and confidence intervals, and gives good-looking output with minimal code. Matplotlib is still needed for fine-tuning.

Q3. Which type of chart would you choose to compare sales across different regions? Justify your answer.

A **bar chart** (vertical, or horizontal if names are long). Regions are discrete categories and sales is a numeric total, so bar length gives an accurate comparison since the eye compares lengths against a common zero baseline easily. Bars can be sorted in descending order and annotated with values, and the chart stays clean. Pie charts are harder to compare when slices are similar, and line charts imply a continuous order that regions do not have.

Q4. What information can be obtained from a histogram and a box plot?

Histogram: shape of the distribution of one numerical variable (symmetric, skewed, bimodal), the central tendency, spread, gaps and approximate frequency in each bin. **Box plot:** the five-number summary (minimum, Q1, median, Q3, maximum), the interquartile range (IQR), skewness through median position and whisker lengths, and outliers beyond 1.5 x IQR. Box plots are especially good for comparing several groups side by side, while histograms show the detailed shape of one distribution.

Q5. Explain the purpose of a scatter plot. How does it help in identifying relationships between variables?

A scatter plot plots each observation as a point using two numerical variables (x and y). It reveals the **direction** (positive or negative), **strength** (tight or scattered points), **form** (linear, curved, clusters) of the relationship, as well as outliers. For example, in Figure 5 the points trend downward, showing profit falls as discount increases. Colour or size can add a third variable. It also helps decide if a linear model or a correlation measure is appropriate. Note that correlation seen in a scatter plot does not prove causation.

Q6. What is a correlation heatmap? How can it assist in feature selection for machine learning?

A correlation heatmap is a colour-coded matrix of pairwise correlation coefficients (-1 to +1) between numerical features, usually from `df.corr()` drawn with `sns.heatmap()`. For feature selection it helps to: (1) pick features strongly correlated with the target, (2) find highly correlated feature pairs (multicollinearity, say |r| > 0.8) and drop one of them to reduce redundancy and overfitting, (3) discard features with near-zero correlation to the target, and (4) understand the data structure quickly. It only captures linear relationships, so non-linear dependence needs other methods.

Q7. Why is chart customization (titles, labels, legends, colors) important in data visualization?

A chart should be understandable without extra explanation. A **title** states what is shown; **axis labels** with units tell what is measured; a **legend** identifies series or categories; **colours** separate groups, highlight important data and, if chosen well (colour-blind friendly, sensible contrast), improve readability; **annotations** draw attention to key points. Good customization reduces misinterpretation, makes the chart professional and helps the audience grasp the message quickly.

Q8. What factors should be considered while selecting an appropriate visualization for a dataset?

- **Type of data:** categorical, numerical, time-series, geographical.
- **Analytical objective:** comparison (bar), trend (line), distribution (histogram, box), relationship (scatter), composition (stacked bar, pie), correlation (heatmap).
- **Number of variables:** univariate, bivariate or multivariate.
- **Data size:** very large data may need density plots or sampling rather than raw scatter.
- **Audience:** technical audiences accept box plots; business audiences prefer simple bar and line charts.
- **Clarity and honesty:** proper scales, no clutter, labelled axes, no distortion.
- **Medium:** printed report, presentation or interactive dashboard.

Q9. How can misleading visualizations affect business decision-making? Give a real-world example.

Misleading charts can cause wrong conclusions, poor investment, wrong budgeting and loss of trust. Common tricks include truncated y-axes, inconsistent scales, 3D effects, cherry-picked time ranges, and pie charts that do not total 100%. **Example:** in 2015, during a US congressional hearing on Planned Parenthood, a chart showed two lines (abortions rising, cancer screenings falling) plotted on the same graph without a shared y-axis scale, so the lines crossed even though the actual numbers were very different (about 327,000 vs 935,000). The chart gave a false impression. In business, a bar chart starting at 95 instead of 0 can make a 2% sales rise look like a huge jump, leading managers to overestimate growth.

Q10. Explain how data visualization supports storytelling and business intelligence in organizations.

In **storytelling**, visuals give data a narrative of context (what happened), conflict (the problem or insight) and resolution (the recommended action). Highlighting the key point, simple design and a logical sequence of charts lead the audience to a conclusion. In **business intelligence**, tools such as Power BI, Tableau and Looker present KPIs on interactive dashboards with drill-down and filters, enabling real-time monitoring of sales, costs and customers. This lets all levels of management spot problems early, compare performance against targets, forecast, and take decisions on the basis of data rather than intuition.

Result: Various visualizations were successfully created and analyzed, and business insights were drawn from them.